

POLITECNICO
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BARIA Motor - Flipped Classroom



Mario Guida
Matteo Mafrici
Martina Lucia Magarelli
Margherita Palitta

Course of Space Propulsion
School of Industrial Engineering
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1 Experimental Data Reduction

The dataset consists of 27 pressure traces acquired at 1 kHz ($\Delta t = 1$ ms) from nine nominally identical propellant batches (pbar2438–pbar2446, except for pbar2444 where the mid and high pressure data were reversed), each fired at three pressure levels by varying the nozzle throat diameter ($D_t = 28.80, 25.26, 21.81$ mm for low, mid, and high pressure, respectively). Data reduction follows the Bayern Chemie (BC) method, which defines two nested time intervals entirely from the trace itself: an outer *action* interval $[t_A, t_G]$ enclosing the complete energetic event, and an inner *burning* interval $[t_B, t_E]$ restricted to the quasi-steady combustion phase.

1.1 Peak Detection

For each trace $p(t)$, the global maximum $p_{\max}$ is located. A 500-sample guard window ($t > 0.5$ s) is enforced: if the detected maximum falls earlier, the search is restarted on the remaining trace and a warning is issued. This prevents ignition overpressure spikes from being mistaken for the quasi-steady peak.

1.2 Action Interval $[t_A, t_G]$ and Threshold Tuning

Following the Bayern-Chemie method, the action interval is bounded by the 5 % threshold crossings of $p_{\max}$:

$$p(t_A) = p(t_G) = 0.05 p_{\max}. \quad (1)$$

A threshold of 5 % is adopted in place of the 1 % value specified in the BC document: for the present dataset the pre-ignition noise floor (≈ 5 bar) exceeds both values on mid- and high-pressure regimes, so the choice has no practical effect; the higher value is retained for future batches, where a lower noise floor would otherwise risk placing t_A inside the ignition transient. For the same forward-compatibility reason, a derivative-stability guard is appended. The raw trace is smoothed with a 30-sample (30 ms) moving average, its derivative $|\dot{p}|$ is computed, and the normalised local standard deviation

$$\delta(t) = \frac{\sigma_{30}(|\dot{p}|)}{\max |\dot{p}|} \quad (2)$$

is used as a stability criterion: $\delta \approx 0$ signals a monotone pressure ramp, $\delta \gg 0$ flags the ignition transient or the tail-off curvature. t_A is placed at the first sample with $\delta < 0.12$ after the 5 % crossing on the rising flank; t_G symmetrically at the last such sample before the crossing on the decaying flank. On the present 27 firings the guard activates on 7 of the 9 low-pressure traces, shifting t_A by up to 47 ms; it has no effect on mid- and high-pressure regimes, where the noise floor activates the threshold before the ignition transient begins.

Table 1: Sensitivity of Vieillé parameters, c^* , and stability-guard behaviour to window size.

	20 ms	30 ms	50 ms
a [mm s ⁻¹ bar ⁻ⁿ]	1.732 ± 0.019	1.744 ± 0.021	1.733 ± 0.022
n [-]	0.3816 ± 0.0028	0.3799 ± 0.0030	0.3813 ± 0.0032
c^* [m s ⁻¹]	1512.1 ± 11.8	1511.6 ± 11.5	1512.3 ± 11.8
t_A guard activ.	10	7	1
t_G guard activ.	0	0	2
Max t_A shift	33 ms	47 ms	68 ms

Visual inspection of the pressure traces and t_A markers suggests that these activations are consistent with inter-batch variability in ignition dynamics rather than a numerical artefact. A sensitivity scan at window sizes of 20, 30, and 50 ms (Table 1) shows that $\Delta a < \sigma_a$, $\Delta n < \sigma_n$, and $\Delta c^* \ll \sigma_{c^*}$ across all windows. The extreme values exhibit physically interpretable failure modes: at 20 ms δ becomes sensitive to brief local fluctuations on the rising edge, producing additional t_A activations on several mid- and high-pressure traces (pbar2440 mid-high, pbar2441 mid, pbar2442 mid, pbar2444 mid, pbar2445 mid) not observed with the 30 ms window; at 50 ms the window retains plateau memory into the tail-off, artificially elevating δ and triggering spurious t_G corrections on pbar2439 and pbar2446 low. The 30 ms window produces the largest t_A shift among the

non-pathological windows, placing t_A most conservatively beyond the ignition transient, and is therefore adopted as the working default.

1.3 Reference Pressure and Burning Interval

The reference pressure is:

$$p_{\text{ref}} = \frac{1}{2 \Delta t_{\text{action}}} \int_{t_A}^{t_G} p(t) dt, \quad \Delta t_{\text{action}} = t_G - t_A, \quad (3)$$

evaluated with the trapezoidal rule; the factor of $1/2$ is part of the BC method definition [1] and places p_{ref} near the midpoint of the quasi-steady plateau. The burning interval $[t_B, t_E]$ is the maximal contiguous region satisfying $p(t) \geq p_{\text{ref}}$ within $[t_A, t_G]$, so that both transients are excluded by construction. The effective pressure is the time-average over the burning interval:

$$P_{\text{eff}} = \frac{1}{t_b} \int_{t_B}^{t_E} p(t) dt, \quad t_b = t_E - t_B. \quad (4)$$

1.4 Burning Rate and Characteristic Velocity

The burning rate follows the TOT definition [2]: $r_b = w_b/t_b$,

where $w_b = 30$ mm is the nominal web thickness (constant for all firings). Restricting t_b to $[t_B, t_E]$ rather than the full action time ensures that numerator and denominator refer to the same quasi-steady event, bounding the non-instantaneous burnout error identified in [2]. The characteristic velocity is derived from each trace as:

$$c^* = \frac{A_t}{m_{\text{prop}}} \int_{t_B}^{t_E} p(t) dt, \quad (5)$$

with $A_t = \pi D_t^2/4$ and m_{prop} the loaded propellant mass. Integration over $[t_B, t_E]$ excludes igniter mass flow and tail-off blowdown, which would otherwise bias c^* low. The formulation assumes a choked, non-eroding nozzle and negligible chamber gas storage, both satisfied for the BARIA motor at the pressure levels tested.

2 Results assessment

2.1 Pre-regime Statistics

Table 2: Experimental results per pressure regime ($N = 9$ each).

Regime	$\bar{P}_{\text{eff}}$ [bar]	CV_P [%]	$\bar{r}_b$ [mm/s]	CV_r [%]	$\bar{t}_b$ [s]	σ_{t_b} [s]
Low ($D_t = 28.80$ mm)	31.60	0.72	6.493	0.64	4.621	0.0311
Mid ($D_t = 25.26$ mm)	47.60	0.97	7.527	0.78	3.986	0.0274
High ($D_t = 21.81$ mm)	75.55	0.88	9.037	0.75	3.320	0.0249

All regimes exhibit CV below 1 %, confirming good batch-to-batch repeatability of the manufacturing process.

2.2 Vieille's Law & Characteristic Velocity

The 27 (P_{eff}, r_b) couples are fitted to: $r_b = a P_{\text{eff}}^n$.

By linearizing in log-log space and applying ordinary least squares to: $\ln r_b = \ln a + n \ln P_{\text{eff}}$. Uncertainty intervals are from the Monte Carlo propagation of Section 2.3:

$$a = 1.744 \pm 0.021 \text{ mm s}^{-1} \text{ bar}^{-n}, \quad n = 0.380 \pm 0.003, \quad R^2 = 0.9984. \quad (6)$$

Averaging the 27 values from Eq. (5): $c^* = 1511.6 \pm 11.5$ m/s (CV = 0.76 %).

The low CV across three throat sizes confirms pressure-independence of c^* , consistent with complete and stable combustion.

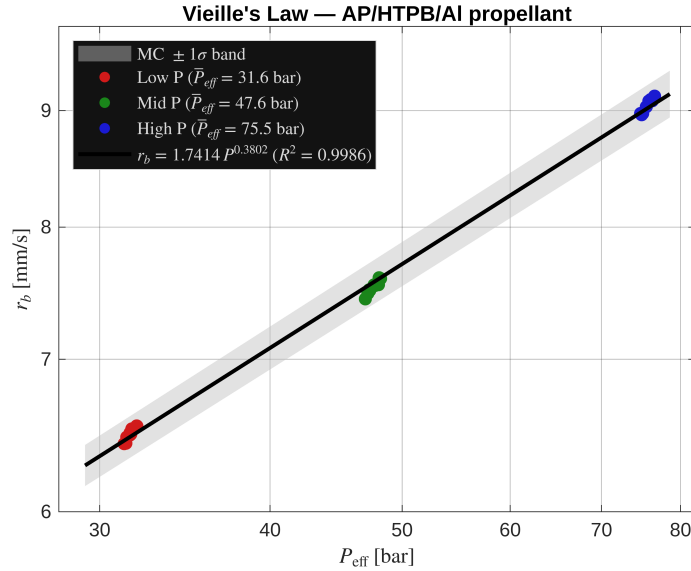


Figure 1: Vieille’s law fit on the 27 (P_{eff} , r_b) couples in log–log scale. The shaded band is the MC $\pm 1\sigma$ uncertainty on a and n .

2.3 Uncertainty Analysis

Uncertainty in a , n , and c^* is propagated via Monte Carlo ($N = 5000$, fixed seed). The OLS regression residuals from the Vieille fit supply the initial estimates σ_a and σ_n , which the MC then propagates through the 0D model (7) to yield distributions of t_b and P_{eff} . Parameters a and n are treated as *aleatoric* (intrinsic propellant variability): sampled as independent Gaussians with means and standard deviations from the OLS regression. The characteristic velocity c^* is treated as *epistemic* (measurement uncertainty): sampled with mean and standard deviation from the 27 experimental values. At each iteration, the 0D model (7) is integrated for the triple (a_i, n_i, c_i^*) , yielding distributions of t_b and P_{eff} . All three configurations converge within $N = 5000$ draws. The MC uncertainty CV (2.6–3.1 %) is approximately four times the experimental batch-to-batch CV ($< 1\%$), confirming that inter-batch scatter reflects genuine propellant variability rather than instrumentation noise.

Table 3: Monte Carlo uncertainty propagation ($N = 5000$).

Regime	Low	Mid	High
σ_{t_b} [s]	0.120	0.108	0.095
CV $_{t_b}$ [%]	2.58	2.71	2.88
σ_P [bar]	0.88	1.40	2.37
CV $_P$ [%]	2.82	2.95	3.10

2.4 Ballistic Model Validation

Quasi-steady internal-ballistics model evaluates the equilibrium chamber pressure at each time step:

$$p = \left(\frac{a c^* \rho_p A_b(t)}{A_t} \right)^{1/(1-n)}, \quad (7)$$

The coefficient a , fitted in $[\text{mm s}^{-1} \text{bar}^{-n}]$, is converted to SI units as $a_{\text{SI}} = a \times 10^{-3}/(10^5)^n$ before insertion into Eq. (7). The model is fed with the fitted Vieille parameters Eq. (6) and c^* Section (2.2); no post-hoc tuning is applied. Validation against mid-pressure firing of pbar2438 ($D_t = 25.26$ mm) gives

$$\text{RMSE}_{[t_B \rightarrow t_E]} = 2.35 \text{ bar} \approx 5\% P_{\text{eff}}. \quad (8)$$

Comparable RMSE values (4–5 % P_{eff}) are observed across all three regimes. The signed error profile suggests a systematic trend: the model tends to underpredict in the first and last quarters of $[t_B, t_E]$ and to overpredict near mid-burn. This pattern is broadly consistent with the Hump effect [3], which arises from spatial variation of r_b within the cured grain driven by propellant rheology, ingredient migration, and casting geometry — none of which are captured by a spatially uniform quasi-steady model. The ± 3 –4 % Hump amplitudes reported in [3] are of the same order as the observed 5 % RMSE.

References

- [1] Politecnico di Milano – Space Propulsion. *The Bayern Chemie Method (BC): Flipped Class on SRM Internal Ballistics*. Course material. Document BC-V01-4. Politecnico di Milano, 2025.
- [2] R.S. Fry et al. “Solid Propellant Burning Rate Measurement Methods Used Within the NATO Propulsion Community”. In: *37th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit*. AIAA 2001-3948. Salt Lake City, UT, July 2001. DOI: 10.2514/6.2001-3948.
- [3] Thomas E. Kallmeyer and Larry H. Sayer. “Differences Between Actual and Predicted Pressure-Time Histories of Solid Rocket Motors”. In: *AIAA/SAE/ASME 18th Joint Propulsion Conference*. AIAA-82-1094. Cleveland, OH, June 1982. DOI: 10.2514/6.1982-1094.